A Machine Learning Approach to Urban Growth Prediction of Juba, South Sudan (2024-2033)

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Abstract

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Chapter 1: Introduction

Chapter Summary

Urbanization in Juba, South Sudan, presents opportunities and challenges, straining infrastructure and environmental sustainability. Rapid, unplanned growth, driven by migration and weak governance, has led to informal settlements and resource shortages. This study uses machine learning (Random Forest) and geospatial analysis to predict urban expansion (2024–2033) based on historical Landsat data (1994–2023).

Key objectives include analyzing past trends, developing a predictive model, and assessing its accuracy. The findings will aid policymakers in sustainable planning, aligning with SDG 11 (sustainable cities). By forecasting growth, the study supports infrastructure development, environmental protection, and risk mitigation, ensuring Juba's urbanization is resilient and equitable.

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1.1 Background

Urbanization is a defining characteristic of modern development, particularly in rapidly expanding cities within developing nations, like Juba in South Sudan. The process of urban growth introduces both opportunities and challenges, shaping the trajectory of economic and social progress (Justin, 2017; Justin and and De Vries, 2019).

In regions such as South Sudan, the pace of urbanization is accelerated by a combination of population growth, migration trends, and economic transitions. While this expansion fosters economic opportunities and infrastructural advancements, it also places considerable strain on essential resources, including housing, transportation networks, public utilities, and environmental sustainability (Zeng *et al.*, 2016; Jiang and Lv, 2022).

In many developing countries, the challenges associated with urban expansion are compounded by weak governance structures, fragile institutional frameworks, and a lack of comprehensive urban planning mechanisms. These factors contribute to the unregulated sprawl of cities, making it increasingly difficult to implement effective development strategies(Zeng *et al.*, 2016; Justin, 2017; Wild, Jok and Patel, 2018; Jiang and Lv, 2022).

Addressing these complexities requires a multi-faceted approach that integrates sustainability principles while simultaneously tackling pressing concerns such as affordable housing, efficient public transportation, equitable resource distribution, and social inclusion. Aligning urban development strategies with global frameworks, particularly the United Nations Sustainable Development Goals (SDGs), is crucial in fostering long-term sustainability. In this context, SDG 11 (sustainable cities and communities), SDG 6 (clean water and sanitation), and SDG 13 (climate action) serve as guiding principles for ensuring balanced urban expansion(Zeng *et al.*, 2016; Maqbool *et al.*, 2020).

Historical Context of South Sudan's Urban Growth

The urbanization trajectory of South Sudan is deeply intertwined with its socio-political history. Decades of conflict and political instability have shaped the nation's development, culminating in its secession from Sudan. On July 9, 2011, South Sudan officially became an independent nation following a referendum, making it the world's newest sovereign state and Africa's 54th country. This historic event marked the beginning of a new chapter for the nation, one defined by both aspirations for progress and the daunting task of post-conflict reconstruction (UN-HABITAT *et al.*, 2023). As a newly established country emerging from prolonged civil wars, South Sudan has faced significant hurdles in building its economy, infrastructure, and governance structures. The post-independence period has been characterized by efforts to stabilize the nation, foster economic growth, and rebuild critical institutions. However, these ambitions have been met with persistent challenges, including ongoing political tensions, economic instability, and weak governance capacity (UN-HABITAT *et al.*, 2023).

Juba: South Sudan's Epicenter of Rapid Urbanization

Juba, the capital and largest city of South Sudan, has been at the forefront of the country's urban expansion, experiencing substantial demographic and infrastructural transformations (Justin, 2017; Justin and De Vries, 2019; Salvatore *et al.*, 2023). The city's growth has been driven by multiple interrelated factors, including rural-to-urban migration, economic activities, and large-scale reconstruction efforts following years of civil conflict. After South Sudan gained independence, Juba became a major destination for displaced populations, returning refugees, and migrants seeking employment, security, and access to essential services (UN-HABITAT *et al.*, 2023). The influx of people into Juba has led to a surge in housing demand, transforming the city's spatial landscape. Previously undeveloped areas and agricultural lands are being rapidly converted into residential, commercial, and industrial zones to accommodate the growing population (UN-HABITAT *et al.*, 2023). However, this expansion has largely been unplanned, resulting in sprawling informal settlements that lack adequate infrastructure and basic services. The unregulated nature of urban expansion has created significant challenges for city planners, including inefficient land use, encroachments on environmentally sensitive areas, and the proliferation of slums (Justin and and De Vries, 2019; UN-HABITAT *et al.*, 2023).

Infrastructure Struggles and Urban Challenges

As Juba continues to expand, its infrastructure development has struggled to keep pace with the growing demands of its residents. The city faces critical deficits in essential urban services, including roads, housing, sanitation systems, and public transportation. Many neighborhoods experience chronic congestion due to poorly maintained road networks and inadequate transportation options. The lack of a reliable waste management system has exacerbated environmental pollution, while deficiencies in water supply and electricity distribution remain major concerns for both residents and businesses (UN-HABITAT *et al.*, 2023).

Furthermore, Juba's evolving urban environment reflects stark contrasts between modern development and persistent infrastructural deficits. High-rise buildings, commercial hubs, and new business centers are emerging alongside informal settlements, highlighting the disparities in urban planning and resource distribution. The rapid transformation of Juba's cityscape underscores both the opportunities and challenges associated with urban growth. On one hand, the city's expansion presents prospects for economic development, increased investments, and improved public services. On the other hand, the lack of coordinated planning and sustainable urban policies threatens long-term resilience and livability (UN-HABITAT *et al.*, 2023).

Therefore, Juba's rapid urbanization calls for urgent need for comprehensive and sustainable urban planning. As the city continues to grow, it is imperative to implement policies that balance development with environmental conservation, social equity, and infrastructural efficiency. Addressing these challenges requires a collaborative approach involving government agencies, urban planners, and international development organizations. By integrating modern urban planning techniques, investing in resilient infrastructure, and aligning growth strategies with global sustainability frameworks, Juba can navigate its urbanization trajectory in a manner that promotes long-term prosperity and stability

1.2 Problem Statement

Urban growth, as conceptualized by the United Nations Human Settlements Program (UN-Habitat), refers to the progressive expansion of towns and cities in terms of both spatial extent and population size (UN-HABITAT, 2006). This dynamic phenomenon unfolds through two primary mechanisms: inward growth, which is characterized by an increase in population density and intensified land use within pre-existing urban boundaries, and outward growth, which involves the physical expansion of urban areas into surrounding rural or undeveloped regions. These growth patterns are particularly pronounced in rapidly urbanizing regions, where economic, social, and demographic factors collectively contribute to the transformation of urban landscapes (Sexton *et al.*, 2013). However, urban growth is a multifaceted process that extends beyond simple expansion, significantly influencing both human settlements and the environment at large (Munthali and Murayama, 2011; Khawaldah, Farhan and Alzboun, 2020).

Urban growth and land use changes

As urban areas expand, they inevitably encroach upon previously undeveloped lands, leading to substantial alterations in Land Use and Land Cover (LULC). This transformation, broadly referred to as Land Use and Land Cover Change (LULCC), denotes modifications to the Earth's surface brought about by both human activities and natural processes over time and space (Rodríguez-Cuenca, Alonso and Tamés-Noriega, 2015). The driving forces behind LULCC include the growing demand for residential, commercial, and infrastructural development, which often results in the conversion of agricultural land and natural ecosystems into urbanized spaces.

Research indicates that urban areas across the globe are expanding at an unprecedented pace, with projections forecasting further and more extensive growth in the coming decades (Ishtiaque, Shrestha and Chhetri, 2017). Given the rapid nature of urbanization, the necessity of monitoring and predicting LULCC through change detection techniques has become increasingly crucial. Change detection methodologies facilitate the systematic tracking, analysis, and management of evolving spatial distributions of land resources, enabling informed decision-making for sustainable urban planning (Turner and Ruscher, 1988). Failure to adequately monitor these transformations can lead to haphazard urban expansion, inefficient infrastructure deployment, and heightened susceptibility to environmental hazards such as flooding, pollution, and resource depletion.

Rapid urban expansion in Juba

Juba, the capital city of South Sudan, serves as a prime example of a rapidly urbanizing metropolis experiencing both densification within its core and sprawling expansion into peripheral areas. The surge in population within the city's established boundaries has exerted immense pressure on public infrastructure and essential services, leading to a multitude of urban challenges, including overcrowding, traffic congestion, inadequate sanitation, and public health concerns. Simultaneously, the outward expansion of Juba has resulted in the transformation of surrounding agricultural land, forests, and wetlands into built-up areas, thereby significantly disrupting local ecosystems and biodiversity (Gudo, Deng and Qureshi, 2022a).

One of the most pressing challenges in managing Juba's urban growth is the lack of comprehensive and up-to-date spatial data on LULCC (Gudo, Deng and Qureshi, 2022b; UN-HABITAT *et al.*, 2023). The city's rapid and largely unregulated expansion, coupled with insufficient monitoring systems, presents substantial difficulties for urban planners and policymakers attempting to anticipate and address the implications of land use changes on infrastructure development and environmental sustainability. Consequently, large swathes of agricultural and natural lands are being converted into urban spaces, leading to altered hydrological patterns, exacerbated environmental degradation, and increased pollution levels(Gudo, Deng and Qureshi, 2022b).

Challenges in urban monitoring and sustainable development

Traditional methods of monitoring urban growth, such as manual land surveys and historical satellite imagery analysis, often fall short in capturing the complexities associated with rapid urbanization, particularly in cities characterized by informal settlements and unregulated construction. The limitations inherent in these conventional approaches underscore the growing necessity for more sophisticated change detection methodologies and geospatial analytical techniques to effectively map and predict urban expansion patterns (Ishtiaque, Shrestha and Chhetri, 2017; Khawaldah, Farhan and Alzboun, 2020) .

Emerging technologies, such as machine learning, remote sensing, and Geographic Information Systems (GIS), have proven to be invaluable tools for urban planners in their efforts to assess real-time land cover changes, detect informal settlements, and identify infrastructural deficits. The integration of these advanced analytical techniques can significantly enhance the efficiency of land-use planning, risk assessment, and environmental management, ensuring that urban expansion is conducted in a manner that is both sustainable and adaptive to future challenges.

Environmental and social-economic challenges

The unregulated and unchecked expansion of urban areas presents a host of human and ecological challenges. On the human side, the continuous influx of people into urban environments necessitates the provision of adequate housing, efficient transportation networks, and robust social services, all of which must evolve in tandem with growing demand. Failure to adequately address these needs can precipitate housing shortages, inefficient public transportation systems, and widening socio-economic disparities.

From an environmental perspective, rapid urbanization results in the depletion of natural resources, habitat loss, deforestation, increased surface temperatures, and alterations in local climate patterns. Additionally, the large-scale conversion of natural landscapes into impervious urban surfaces has far-reaching consequences for hydrological systems, disrupting water cycles, exacerbating flood risks, and contributing to groundwater depletion. These ecological disruptions pose serious long-term sustainability challenges, emphasizing the urgency of adopting urban planning strategies that strike a balance between developmental needs and environmental conservation (Munthali and Murayama, 2011; Sexton *et al.*, 2013; Munthali *et al.*, 2019; Khawaldah, Farhan and Alzboun, 2020).

Summary

Like other rapidly evolving urban cities in developing countries, Juba's ongoing urban expansion can be seen in a two-faced reality: on one hand, it signifies a profound opportunity for economic development, while on the other, it poses significant challenges for sustainable resource management. Rapid urbanization, if unchecked, can lead to issues such as infrastructure strain, environmental degradation, and the displacement of vulnerable populations. Therefore, the effective prediction of urban growth is essential for informed decision-making and sustainable development. To mitigate the adverse consequences related to rapid urbanization, it is crucial to implement integrated urban planning strategies that prioritize the development of resilient infrastructure, efficient land-use policies, and climate-conscious urban designs. Leveraging cutting-edge geospatial technologies, remote sensing data, and predictive analytics can play a pivotal role in this effort.

In this study, we propose the prediction of Juba’s urban growth from 2024 to 2033 based on Random Forest, a powerful and widely used Machine Learning model for spatial prediction tasks (mainly characterized into classification and regression tasks). By analyzing Landsat 5 and 7 data spanning from 1994 to 2023, this approach allows us to predict land-use changes, and urban sprawl with high accuracy. Random Forest excels in handling large, complex datasets and capturing non-linear relationships between urban growth drivers, such as population density, land cover changes, and socioeconomic factors.

When trained on historical land-use patterns, and by recognizing these patterns, the Random Forest model can predict future urban growth trends. These predictions are vital for urban planners and policymakers to better monitor, forecast, and guide Juba's expansion in a manner that fosters economic growth while minimizing environmental and social challenges.

Achieving sustainable urban development in Juba will necessitate collaborative efforts between governmental agencies, academic researchers such as in this study, and local communities, ensuring that urban expansion aligns with broader sustainability objectives and contributes to the overall well-being of its inhabitants. Only through proactive and data-driven urban management approaches, such as this Random Forest-based prediction model, can Juba effectively navigate the complexities of its rapid growth trajectory. This study will contribute to creating resilient, sustainable, and economically viable urban environments, ensuring that Juba can emerge as a model for future cities in the region.

1.3 Objectives of the Study

1.3.1 Primary objective

In this study, we seek to develop a comprehensive understanding of both historical and future urban growth patterns in Juba, South Sudan, by analyzing past trends from 1994 to 2023 and predicting future expansion from 2024 up to the year 2033. This includes utilizing advanced predictive models, specifically Machine Learning's Random Forest and advanced geospatial pre-processing techniques such as cloud masking and ground control points (GCPs) to collect, extract, and manipulate data to inform urban planning and development strategies.

1.3.2 Specific Objectives

To collect historical time series data from Landsat 5 and 7 spanning 1994 to 2023.

To develop a Random Forest machine learning model that identifies historical urban growth trends in Juba from 1994 to 2023, focusing on changes in land use and land cover.

To train the model to generate visible historical urban growth patterns merged for the years 1994-2003, 2004-2013 and 2014-2023.

To evaluate the performance of the trained model (Random Forest) using a confusion matrix as a metric.

To predict future urban growth patterns in Juba for the period 2024-2033 using the trained model.

1.4 Significance of the study

Urbanization in Juba, South Sudan, is rapidly transforming the city and the landscape of its neighborhood, presenting both opportunities for economic development and challenges related to infrastructure, environmental sustainability, and resource management. Understanding past trends and forecasting future urban expansion will be vital for informed decision-making and sustainable urban planning.

This study is significant as it leverages Machine Learning's Random Forest and advanced open-source geospatial pre-processing techniques using both Google Earth Engine (GEE) and Quantum Geographic Information System (GIS), including cloud masking and ground control points (GCPs), to analyze urban growth patterns. By utilizing Landsat 5 and 7 data from 1994 to 2023, this research provides an Informatics-informed data-driven foundation for predicting Juba's urban expansion from 2024 to 2033. The findings will enable urban planners, policymakers, and researchers to:

Enhance Urban Planning: By identifying historical land-use trends and forecasting future urban growth, national, state, and city planners and policy makers can implement data-driven policies that promote sustainable development and efficient land management.

Support Infrastructure Development: Accurate predictions of Juba's urban expansion can inform the strategic placement of roads, utilities, housing, disaster preparedness, and public services, ensuring that infrastructure development aligns with population growth.

Mitigate Environmental and Social Risks: Understanding patterns of urbanization allows stakeholders to anticipate and address potential challenges, such as deforestation, loss of green spaces, flood planning, and informal settlements, which can impact climate resilience and socio-economic stability.

Advance Geospatial and Machine Learning Applications: This study contributes to the growing field of geospatial machine learning by demonstrating the effectiveness of Random Forest in modeling and predicting urban growth, setting a benchmark for future studies in the region.

Provide a Decision-Support Tool: The predictive model developed in this study can serve as a decision-support system, enabling government agencies, urban developers, and policymakers to make proactive, evidence-based choices for Juba’s future growth.

By combining historical analysis with predictive modeling, this research will not only deepen the understanding of Juba’s urban transformation but also provide practical insights for managing its expansion in a sustainable and resilient manner.

1.5 Research Questions

To achieve the research objectives, the study shall address four (4) key questions. These are:

1. What are the historical trends and patterns of urban growth in Juba from 1994 to the present (2023) based on Landsat 5 and 7 data?
2. How well does the trained Random Forest model perform in detecting historical urban growth trends in Juba from 1994-2003?
3. What are the anticipated (predicted) future patterns of urban growth in Juba from 2024-2033?
4. How can predictive urban growth modeling support decision-making for sustainable land management and infrastructure planning in Juba?

Chapter 2: Literature review

Chapter Summary

This literature review examines urban growth prediction methodologies, focusing on Juba, South Sudan - a rapidly urbanizing post-conflict city. Key approaches include Random Forest (RF) machine learning for land cover classification, Cellular Automata for spatial dynamics, and Markov Chains for transition probabilities.

The study highlights Juba's unique challenges: scarce data due to conflict, unplanned expansion, and environmental impacts. By integrating Google Earth Engine and QGIS, this research advances urban prediction in data-scarce contexts using 1994-2023 Landsat imagery.

The novelty lies in applying RF with cloud masking and ground control points to develop a decision-support tool for sustainable planning in fragile urban environments.

2.1 Introduction

Urban growth prediction is an important component of urban planning and sustainable development, particularly in rapidly urbanizing regions such as Juba, the capital of Central Equatoria State that doubles as the Capital city of South Sudan. Juba has experienced significant changes in land use and land cover (LULC) over the past few decades as noted by (Gudo, Deng and Qureshi, 2022a; UN-HABITAT *et al.*, 2023), driven by factors such as population growth, civil unrest, and economic development. This literature review explores the methodologies and findings of recent studies on urban growth prediction in general and Juba in particular.

2.2 Methodologies for Urban Growth Prediction

Urban growth prediction methodologies employ a variety of models and technologies to simulate future urban expansion patterns. These approaches aim to generate accurate forecasts that support urban planning, infrastructure development, and policy-making, ensuring sustainable urban growth. By analyzing historical trends and incorporating spatial and statistical data, these methodologies provide insights into the drivers of urbanization and help stakeholders make informed decisions.

2.2.1 Machine Learning Models (Random Forest)

Machine learning models have become indispensable tools in urban growth prediction, offering the capability to analyze complex patterns in large spatial datasets. Among these, Random Forest (RF) has gained significant traction for land use classification due to its ability to leverage historical remote sensing data and spatial attributes to identify urban expansion trends (Balzter *et al.*, 2015; Ouma *et al.*, 2024; Sulistiyono *et al.*, 2024). Similarly, Support Vector Machines (SVM) have proven effective in distinguishing land cover types and detecting urbanization dynamics by identifying decision boundaries in high-dimensional feature spaces (Bouaziz, Eisold and Guermazi, 2017; Trabelsi *et al.*, 2023). Meanwhile, Artificial Neural Networks (ANNs), with their capacity to model nonlinear relationships, have been increasingly employed to enhance predictive accuracy in urban growth forecasting (Feng *et al.*, 2011).

Ensemble learning techniques, particularly Random Forest, have been instrumental in improving predictive performance and robustness in urban expansion modeling. Introduced by Breiman (2001), Random Forest is a powerful ensemble method that constructs multiple decision trees during training, each trained on a random subset of the dataset (Breiman, 2001). This bagging (bootstrap aggregating) approach reduces model variance and enhances generalizability, making RF especially effective for noisy or high-dimensional datasets. Unlike traditional decision trees, which are prone to overfitting, RF leverages randomness at multiple stages—both in bootstrap sampling and feature selection—to generate de-correlated trees. At each node, only a subset of features is considered for splitting, allowing the model to uncover complex interactions between spatial and socio-economic factors influencing urbanization (Balzter *et al.*, 2015; Sulistiyono *et al.*, 2024).

A key advantage of RF is its built-in mechanism for estimating model accuracy through Out-of-Bag (OOB) error. In each iteration, approximately one-third of the data is left out of the bootstrap sample, allowing for internal cross-validation without the need for a separate test set (Sulistiyono *et al.*, 2024). This feature makes RF a robust and efficient choice for LULCC modeling, particularly in scenarios where data availability is limited (Breiman, 2001). Given its ability to capture spatial heterogeneity and account for uncertainties, Random Forest remains a preferred machine learning model for urban growth prediction, aiding policymakers and urban planners in making data-driven decisions for sustainable development.

2.2.2 Cellular Automata (CA)

Cellular Automata (CA) models are widely used in Land Use and Land Cover Change (LULCC) studies due to their ability to simulate spatial dynamics through local interactions. CA is a mathematical modeling approach that represents landscapes as grids of interacting cells, where each cell's state evolves based on predefined transition rules and the states of its neighboring cells. This method effectively captures the self-organizing behavior of urban systems, producing self-similar patterns and highlighting both spatial and quantitative changes over time (Khawaldah, Farhan and Alzboun, 2020; Das and Angadi, 2022).

In the context of urban expansion, CA models simulate land use changes by integrating various influencing factors such as population density, infrastructure development, road networks, and exclusion zones. These models are particularly effective in capturing the nonlinear nature of urban growth, where small-scale local interactions contribute to large-scale spatial transformations (Khawaldah, Farhan and Alzboun, 2020; Das and Angadi, 2022).

Variants of CA, such as Logistic Regression–Cellular Automata (LR-CA) and Artificial Neural Network–Cellular Automata (ANN-CA), enhance prediction accuracy by incorporating statistical and deep learning techniques (Feng *et al.*, 2011) (Feng et al., 2011; Justin and De Vries, 2019). By leveraging historical land cover changes and spatial constraints, these hybrid models refine urban growth simulations and provide more accurate projections for future development.

CA-based models have been successfully applied to predict urban growth patterns in various cities, demonstrating their ability to replicate real-world spatial dynamics. Their adaptability and capacity to integrate diverse data sources make them valuable tools in urban planning and LULCC studies.

2.2.3 Markov Chain Models

Markov Chain models are widely utilized for predicting land-use and land-cover change (LULCC) by estimating the probability of transition from one land-use type to another over a given time period. These models analyze historical land-use data to determine the likelihood of changes occurring in the future, making them particularly effective for assessing long-term urban growth patterns (Al-Hameedi *et al.*, 2021). By computing transition matrices, Markov Chain models can identify areas with a high probability of urban expansion and forecast potential land-use distributions.

One of the key strengths of Markov Chain models is their ability to provide quantitative insights into urbanization trends, allowing planners to anticipate spatial transformations and make informed policy decisions (Das and Angadi, 2022). However, because Markov models rely solely on historical transition probabilities, they do not inherently account for spatial constraints, environmental factors, or socioeconomic influences. To address this limitation, they are often integrated with Cellular Automata (CA) models, which introduce spatial dynamics by simulating interactions between neighboring land parcels (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021; Das and Angadi, 2022). This hybrid approach enhances the predictive accuracy of urban growth models by incorporating both probabilistic and spatial elements.

Markov Chain models have been successfully applied in various urban settings to forecast future expansion scenarios and assess land-use sustainability. Studies have demonstrated their effectiveness in identifying high-risk zones for urban sprawl, enabling proactive planning interventions to mitigate adverse environmental impacts (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021; Das and Angadi, 2022). By simulating different urban growth scenarios, these models contribute to strategic land-use planning, infrastructure development, and environmental conservation efforts, ensuring a more sustainable and resilient approach to urban expansion.

2.2.4 Artificial Neural Networks (ANN)

Artificial Neural Networks (ANNs) are advanced computational models inspired by the structure and function of the human brain (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021). They are designed to recognize patterns, process complex data, and make predictions based on input variables. In the context of urban growth prediction, ANNs are used to model intricate relationships between various factors, such as socio-economic conditions, land-use policies, environmental factors, and infrastructure development. Their ability to process large datasets and capture non-linear interactions makes them particularly effective in understanding urban dynamics (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021).

Studies employing ANNs have shown remarkable success in accurately forecasting urban growth patterns. By leveraging Geographic Information System (GIS) data, ANNs can incorporate spatial and temporal elements into their predictions, enhancing the model's reliability. These models are particularly useful in identifying areas that are likely to experience urban expansion, as well as understanding the key drivers of this growth, such as population density, economic development, and policy decisions (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021).

The application of ANNs in urban growth modeling has provided valuable insights into future land-use trends, helping urban planners and policymakers make informed decisions regarding infrastructure development and land-use management. ANNs have proven instrumental in identifying potential urban sprawl areas, understanding the complex factors influencing growth, and assessing the sustainability of urban expansion (Somvanshi *et al.*, 2020; Al-Hameedi *et al.*, 2021). Through their predictive capabilities, ANNs enable the formulation of more effective strategies for managing urbanization, ensuring that cities can grow in a way that is both sustainable and responsive to the needs of their populations.

2.2.5 Google Earth Engine (Interface) and other Models

Geospatial and remote sensing techniques play a crucial role in urban growth prediction by providing accurate land cover data and spatial analysis tools. Satellite imagery from sources such as Landsat and Sentinel enables the detection of land use changes over time through spectral analysis. Google Earth Engine (GEE) facilitates large-scale spatial analysis and geospatial data processing, while Geographic Information Systems (GIS) provide visualization and spatial integration capabilities for urban development assessments (Bruno Soares, Daly and Dessai, 2018; Nyasulu *et al.*, 2022).

Hybrid and ensemble models combine different methodologies to improve predictive accuracy. By integrating machine learning classification techniques with CA-based spatial simulation, these models refine urban growth predictions (Cifuentes *et al.*, 2020; Nyasulu *et al.*, 2022). Techniques such as stacking and weighted averaging utilize multiple predictive models, enhancing overall forecasting reliability and robustness.

Statistical and econometric models also contribute to urban growth prediction. Markov Chain Analysis is frequently used to predict land use transitions by analyzing historical changes and estimating probabilistic shifts in land cover. Geographically Weighted Regression (GWR) accounts for spatial variations in urban growth determinants, providing more localized and detailed predictions (Zhao *et al.*, 2009; Micheal, 2025).

Scenario-based approaches offer valuable insights into potential future urban expansion under different policy frameworks. Urban growth scenarios such as business-as-usual, compact city development, and urban sprawl simulations help planners evaluate the impacts of different urbanization strategies. Additionally, agent-based models (ABM) simulate individual or group decision-making processes, illustrating how various factors influence urban expansion patterns (Dai *et al.*, 2020; Gobin and Van Herzele, 2023).

2.3 Land Use and Land Cover (LULC) Dynamics in Juba

Several studies have examined land use and land cover (LULC) changes in Juba, consistently highlighting the rapid expansion of built-up areas at the expense of agricultural land and natural ecosystems. One study by Gudo et al. focused on the spatiotemporal dynamics of LULC changes in Jubek State, which encompasses Juba. Using Landsat satellite imagery from 2000 to 2017, the study analyzed transformations across six major land cover categories: built-up areas, water bodies, agricultural land, forest land, barren land, and grassland (Gudo, Deng and Qureshi, 2022b).

The findings revealed a substantial increase in built-up areas, particularly between 2009 and 2017, driven by factors such as population growth, rural-to-urban migration, and economic expansion. Urbanization in Juba followed a linear trajectory alongside population growth, with patterns influenced by security conditions, the availability of natural resources, and broader socio-economic and political factors. The prolonged civil unrest in South Sudan further exacerbated these trends, as large numbers of people migrated to Juba in search of safety and economic opportunities. This influx heightened demand for housing, infrastructure, and public services, accelerating the city's spatial expansion.

In addition to documenting urban growth, the study also identified a critical challenge in predicting future expansion: the lack of consistent and reliable data. Political instability and conflict in the region have disrupted systematic data collection efforts, limiting the availability of high-quality geospatial information (Gudo, Deng and Qureshi, 2022b).

These constraints have made it difficult to develop accurate predictive models for urban growth, underscoring the need for improved data acquisition strategies and methodological approaches to enhance urban planning and decision-making in Juba.

2.4 Land use and Land cover change dynamics in Africa

A study conducted by Faisal Koko, Auwalu et al., investigated the patterns of land use and land cover (LULC) changes in Lagos, Nigeria, employing maximum likelihood classification and post-classification change detection techniques (Faisal Koko *et al.*, 2021). These methods were used to analyze the city's urban expansion and the dynamics of land cover transformation over several decades.

The study identified three distinct phases of urban growth in Lagos and revealed a significant increase in the city's built-up area. In 1990, the built-up area covered approximately 496 km², which expanded to around 860 km² by the year 2000. The urban footprint continued to grow, reaching 1,113 km² in 2010 and surpassing 1,256 km² by 2021. This rapid expansion was accompanied by a marked decline in other land cover types, with vegetation cover decreasing by approximately 398 km², water bodies shrinking by around 246 km², and bare soil reducing by 115 km² between 1990 and 2020 (Faisal Koko *et al.*, 2021).

The study highlighted the environmental consequences of these shifts, particularly the heightened risk of urban flooding. The loss of vegetation and water bodies, combined with the increase in impervious surfaces, has disrupted natural drainage systems, exacerbating both natural and human-induced flooding events . Given these findings, the researchers emphasized the need for sustainable urban planning and proactive flood mitigation strategies to address the growing challenges posed by Lagos' rapid urbanization.

Another study by Ouma et al., explored the application of a Random Forest Regression (RFR) model for predicting land use and land cover change (LULCC) in the Gaborone Dam catchment, Botswana, for the period 2019–2030 (Ouma *et al.*, 2024). The study compared the performance of RFR with two other modeling approaches: logistic regression–cellular automata (LR-CA) and artificial neural network–cellular automata (ANN-CA). The researchers proposed RFR as the preferred model due to its ability to capture complex interactions between land use intensity and the nonlinear relationships between various change-driving factors.

To forecast LULCC, the study incorporated a range of influencing variables categorized into physiographic factors, such as elevation, slope, and aspect, and proximity-neighborhood factors, which included distances to water bodies, roads, and urban areas. The historical LULC patterns from 1986 to 2019 were simulated at five-year intervals to assess the performance of each model. The results demonstrated that RFR significantly outperformed ANN-CA and LR-CA, achieving an accuracy of 84.9%, compared to 62.1% and 60.7%, respectively (Ouma *et al.*, 2024).

Using the RFR model, the study projected notable changes in land cover between 2019 and 2030. The analysis indicated a decline in vegetation cover by 8.9%, while bare soil increased by the same margin. Built-up areas expanded by 2.49%, whereas cropland shrank by 2.8%. Water bodies exhibited minimal change; however, the study found a strong correlation between urban expansion and diminishing dam water capacity. This suggests that increasing population density and urbanization may exert further pressure on water resources.

The study's methodology underscores the potential of machine learning models like RFR in understanding the land–water nexus within catchment areas. These insights can be instrumental in formulating effective policies for sustainable catchment management, urban planning, and water resource conservation in regions experiencing rapid environmental and demographic changes.

2.5 Novelty of the Study

This study introduces a novel approach to predicting urban growth in Juba, South Sudan, by integrating advanced machine learning techniques with geospatial technologies. It aims to address the unique challenges of rapid urbanization in a developing, post-conflict urban center. While urban growth prediction models such as Random Forest (RF), Cellular Automata (CA), and Markov Chains have been widely used globally, their application in Juba—a city undergoing rapid, often unplanned expansion due to population growth, civil unrest, and economic pressures—offers a distinct contribution to the field.

The novelty of this research lies in its context-specific application in Juba, a city where consistent geospatial data is scarce due to prolonged political instability and conflict. By leveraging historical Landsat 5 and 7 data from 1994 to 2023, this study pioneers a data-driven approach in a region where urban growth analyses have been limited. This work fills a critical gap in urban planning knowledge for South Sudan, offering a unique perspective on urban growth in a resource-constrained setting.

This research also integrates Random Forest with advanced geospatial pre-processing techniques, such as cloud masking and ground control points (GCPs), using Google Earth Engine (GEE) and Quantum GIS. These methods ensure robust data quality despite challenges such as cloud cover and limited ground-truthing in Juba, setting a new standard for predictive modeling in data-scarce environments. Additionally, the study uses a 29-year historical dataset (1994–2023) to train the Random Forest model, segmented into decadal patterns, and provides a decade-long future prediction (2024–2033), offering urban planners a detailed temporal framework to anticipate and manage growth.

Another key aspect of this study is its focus on sustainability and resilience. Beyond prediction, the study aims to inform integrated urban planning strategies by forecasting land-use changes and urban sprawl with high accuracy. This directly addresses Juba's dual challenges of economic opportunity and sustainability issues, such as infrastructure strain, environmental degradation, and displacement. The study provides actionable insights that align with global sustainability goals while being locally relevant.

Finally, the development of a Random Forest-based predictive model as a decision-support system is particularly innovative for Juba. Unlike studies that focus solely on model accuracy, this research evaluates performance using a confusion matrix and translates predictions into practical tools for policymakers. This approach enhances infrastructure development, disaster preparedness, and environmental risk mitigation, offering a proactive approach to urban planning in a rapidly transforming city.

Chapter 3: Methodology

Chapter Summary

Landsat 5 (1994-2013) and Landsat 7 (2004-2023) satellite imagery data are obtained from Google Earth Engine (GEE) to analyze urban trends of the ROI, encircling Juba. A Random Forest classifier categorizes land cover into Urban, Bare Land, Water, and Vegetation for three periods (1994-2003, 2004-2013, 2014-2023). Future projections (2024-2033) are derived from spectral trends. Accuracy is assessed through Producer's Accuracy (PA), User's Accuracy (UA), and Overall Accuracy from confusion matrices, ensuring reliable classifications. The workflow combines GEE's cloud processing with QGIS visualization, for implementing the methodology

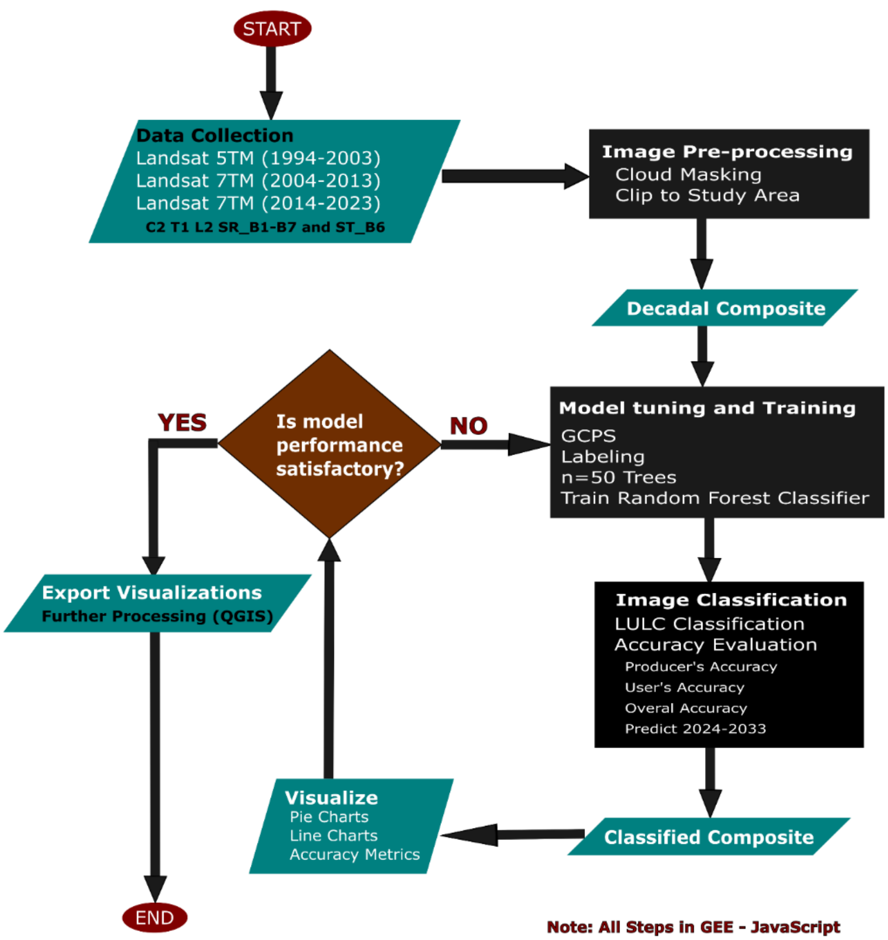


Figure 3-1: Flow chart of the methodology

3.1 Introduction

This study utilizes a machine learning framework to predict urban growth and land cover dynamics in Juba, South Sudan, for the period 2024 to 2033. By leveraging historical satellite imagery from 1994 to 2023, the study applies advanced geospatial analysis techniques to assess past trends and forecast future urban expansion. **Figure 3-1** shows a detailed flow chart of the methodology.

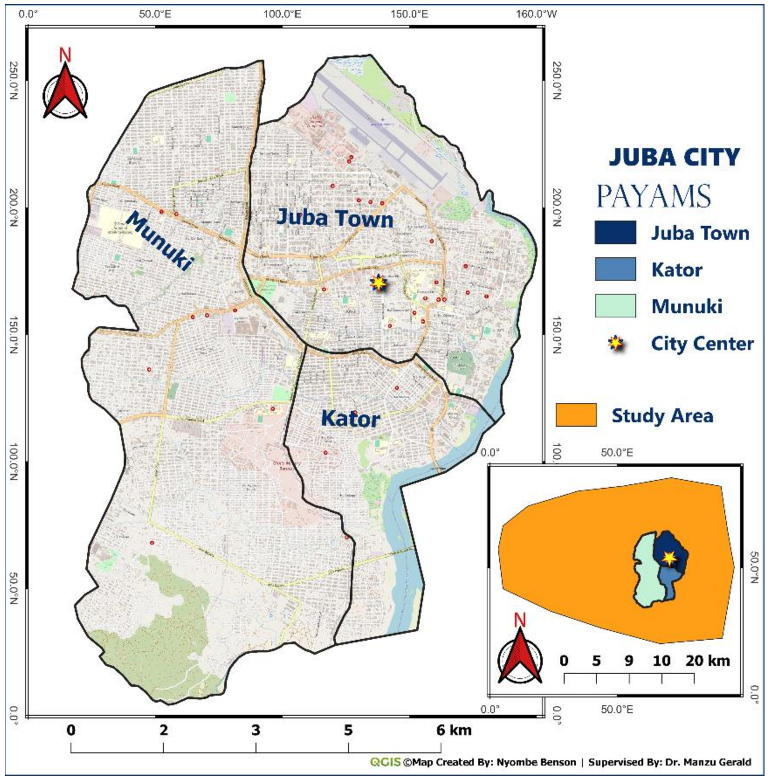
The methodology integrates satellite data processing, machine learning classification, and predictive modeling to generate insights into land-use changes.

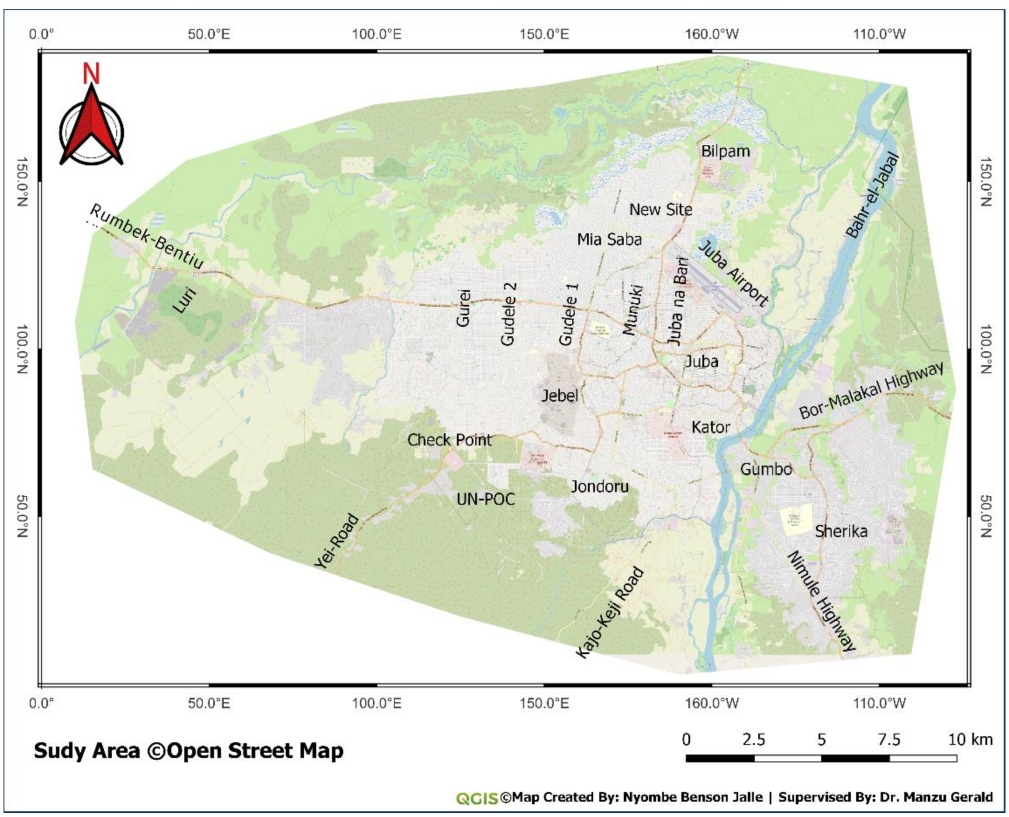
Using Google Earth Engine (GEE) as the primary processing platform, the study employs the Random Forest (RF) algorithm to classify land cover into four categories: Urban, Bare Land, Water, and Vegetation. Historical classifications are generated for three distinct time periods (1994–2003, 2004–2013, and 2014–2023), providing a basis for long-term trend analysis. These classifications inform predictive modeling, which projects future land cover changes based on observed patterns. The study also incorporates spatial analysis and visualization tools, such as Quantum GIS (QGIS), to interpret and present the results. By employing statistical evaluations and validation techniques, the methodology ensures accuracy and reliability in predicting urban expansion. The approach not only enhances understanding of Juba's urbanization dynamics but also serves as a decision-support tool for sustainable urban planning in South Sudan.

3.2 Study Area

The region of interest in the study area extends beyond the three administrative payams of Juba City, namely: Munuki, Kator and Juba Town, to encompass the functional agglomeration of Greater Juba. Historically, Juba City's urban footprint was largely confined within these administrative divisions. However, in recent years, rapid urban expansion driven by population growth, economic activities, and post-conflict reconstruction efforts has led to a significant increase in the city's built-up area. As a result, the urban fabric of Juba has extended well beyond its original administrative boundaries, necessitating a more comprehensive approach to delineating its spatial extent. However, despite the evident expansion of the city, defining the precise borders of the Greater Juba is challenging, due to the absence of an officially established perimeter. For instance, a report by the United Nations Human Settlements Program (UN-HABITAT) in April 2021 highlighted this issue, stating that "The border of Juba municipality is not readily available, which makes it difficult to authoritatively define the area. For example, in 2009, JICA (Japan International Cooperation Agency) estimated the urbanized area of Juba to be around $52km^2$, which is smaller than greater Juba, which UN-OCHA had in 2007 estimated as covering $336km^2$" (UN-HABITAT, 2021). This variation in estimations highlights the inconsistencies in defining Juba's urban extent. As a result, our approach involves delineating a region of interest that, while reflective of the city's functional urban expansion, does not necessarily align with any legally or officially recognized jurisdiction under the designation of "Greater Juba." For the purposes of this study, we shall refer to this delineated area as the "study area" or the "Region of Interest (ROI)".

Given the lack of a standardized definition for Greater Juba's boundaries, the UN-Habitat report (UN-HABITAT, 2021) provided a critical reference for delineating the study area to capture the most extensive representation of Juba's urban landscape. The study area was carefully selected based on visual assessments of digital elevation models and settlement distribution patterns within the city and its surrounding regions. As a result, the designated region of interest encompasses approximately 580.08 km^2, ensuring the inclusion of a comprehensive range of urban characteristics and spatial dynamics.


JUBA CITY
PAYAMS
Juba Town
Kator
Munuki
City Center
Study Area
Juba Town
Munuki
Kator
0 2 3 5 6 km
0 5 9 10 20 km
QGIS ©Map Created By: Nyombe Benson | Supervised By: Dr. Manzu Gerald


Bilpam
New Site
Mia Saba
Rumbek-Bentiu
Luri
Gurei
Gudele 2
Gudele 1
Munuki
Juba na Bari
Juba Airport
Bahr-el-Jabal
Juba
Jebel
Kator
Bor-Malakal Highway
Check Point
Gumbo
UN-POC
Jondoru
Sherika
Yei-Road
Kajo-Keji Road
Nimule Highway
0 2.5 5 7.5 10 km
Sudy Area ©Open Street Map
QGIS ©Map Created By: Nyombe Benson Jalle | Supervised By: Dr. Manzu Gerald

3.3 Data Acquisition and Preprocessing Overview

This study relies on satellite imagery and ground truth points to develop a robust dataset for urban growth prediction in Juba, South Sudan. For 1994-2003, Landsat 5 Collection 2, Tier 1, Level 2 surface reflectance (SR_B1 to SR_B7) and thermal data (ST_B6) are used, while Landsat 7 data of the same tier and level cover 2004-2013 and 2014-2023. Ground Control Points (GCPs) are provided as feature-collections with a landcover property from 0-3 for each class, which define training signatures. The region of interest (ROI) is specified as a geometry asset. Preprocessing ensures data consistency: optical bands are scaled to reflectance using the equation

$$R_{scaled} = \frac{(DN * Multiplicative\ Factor) + Additive\ Offset}{\sin(\theta)}$$

Where:

R_{scaled} is the scaled reflectance value, DN is the raw digital number from the sensor, the $Multiplicative\ Factor$ is 0.0000275 (specific to Landsat Collection 2 Level 2 data), the $Additive\ Offset$ is -0.2, and θ is the solar elevation angle in radians, accounting for solar illumination variations across image acquisition dates.

The Thermal band is converted to kelvin with:

$$T_{kelvin} = (DN * Thermal\ Multiplicative\ Factor) + Thermal\ Additive\ Offset$$

Where:

T_{kelvin} is the temperature in kelvin, the $Thermal\ Multiplicative\ Factor$ is 0.00341802, and the $Thermal\ Additive\ Offset$ is 149.0, converting raw thermal data to a physically meaningful scale.

Clouds and shadows are masked using the QA_PIXEL band, a critical component of Landsat Collection 2 Level 2 data. The QA_PIXEL band is a 16-bit unsigned integer layer that encodes quality information for each pixel in a bit-packed format, where each bit or combination of bits flags specific conditions such as clouds, shadows, water, or snow. For Landsat 5 and 7, this band is generated during atmospheric correction and quality assessment processes, providing a standardized way to identify and filter out contaminated pixels. Each bit position corresponds to a specific condition, and its value (0 or 1) indicates whether that condition is absent or present. In this study, the focus is on masking clouds and cloud shadows, which obscure surface features and could skew classification results. The bit assignments for Landsat Collection 2 Level 2 data are as follows (relevant bits highlighted):

- Bit 0: Fill (1 if data is invalid, e.g., outside the image extent)
- Bit 1: Dilated Cloud (1 if near a cloud, expanding the cloud boundary)
- Bit 2: Cirrus (1 if cirrus clouds are detected, less relevant for Landsat 5/7)
- Bit 3: Cloud (1 if a cloud is present)
- Bit 4: Cloud Shadow (1 if a cloud shadow is present)
- Bits 5-15: Additional flags (e.g., snow, water, confidence levels)

In the masking process, we are only concerned with Bit 3 and Bit 4, which represents clouds and cloud shadows. The process in Google Earth Engine involves extracting the cloud shadow information from the QA_PIXEL band using bitwise operation. Since the QA_PIXEL band encodes multiple quality indicators within a 16-bit integer we use bitwise shifting and logical operations to isolate and remove cloud-affected pixels through:

$$Mask = \left(\frac{QA_{PIXEL}}{2^3} mod\ 2 = 0\right) \wedge \left(\frac{QA\_PIXEL}{2^4} mod\ 2 = 0\right)$$

Where:

- QA_PIXEL is the quality assurance band for each pixel
- 2^3 represents the bit position for cloud (Bit 3)
- 2^4 represents the bit position for cloud shadow (Bit 4)
- The modulo operation, $mod\ 2$ checks whether the bit is set as (1) or not (0)
- $\frac{QA_{PIXEL}}{2^3} mod\ 2 = 0$ ensures the cloud (bit 3) is not set
- $\frac{QA_{PIXEL}}{2^4} mod\ 2 = 0$ ensures the cloud shadow (Bit 4) is not set
- The pixels where both conditions are true (bit is 0) are retained
- $\wedge$ is the logical AND operator

Alternatively, we represent the above equation as:

$$Mask = \big((QA_{PIXEL} \gg 3)\ mod\ 2 = 0\big) \wedge ((QA_{PIXEL} \gg 4)\ mod\ 2 = 0) \wedge$$

Where:

- $\gg$ is the bitwise right shift operator, shifting the relevant bit to the least significant position before applying the modulo operation.

Median composites are then created by computing the median value of unmasked (clear) pixels across the time range for each period, clipped to the ROI to reduce temporal noise. These median composites represent the periods 1994-2003, 2003-2013, and 2014-2023, which are fed for Random Forest training.

3.4 Random Forest Classification

Historical classifications utilize a Random Forest (RF) classifier, an ensemble method that enhances accuracy by combining multiple decision trees.

For a dataset with (n) features (spectral bands SR_B1 to SR_B7 and ST_B6, where
) and (m) training samples (GCPs), RF constructs (T) trees, each trained on a bootstrapped subset of the data and a random subset of features at each node. The final classification for a pixel with feature vector $x = [x_1, x_2, \dots, x_n]$ is determined by majority voting. The classification is represented by:

$$h(x) = mode\{h_1(x), h_2(x), \dots, h_T(x)\}$$

Where:

$x = [x_1, x_2, \dots, x_n]$ is the feature vector for a pixel, consisting of $n = 8$ spectral bands: SR_B1 to SR_B7 and ST_B6.

$h_t(x)$ is the class (0 for Urban, 1 for Bare, 2 for water, or 3 for vegetation) predicted by the *t-th* decision tree in the ensemble.

$T = 50$ is the number of decision trees used in the Random Forest, chosen to balance computational efficiency and robustness.

$mode$ represents the majority voting function, selecting the most frequently predicted class among all trees.

3.4.1 Historical Land Cover Classification

For 1994-2003, the Landsat 5 median composite is classified using the RF model trained on 2003 GCPs (urban2003, bare2003, water2003, vegetation2003), assigning pixels to Urban (0), Bare (1), Water (2), or Vegetation (3) via the majority vote from 50 trees. Validation employs a confusion matrix, with overall accuracy calculated as:

$$Overal\ Accuracy = \frac{\sum_{i=0}^{3} True\ Positives_i}{Total\ Number\ of\ Pixels\ in\ Validation\ Set}$$

Where:

- $True\ Positives_i$ is the number of pixels correctly classified as class (i) in a withheld validation subset (for instance 20% of GCPs),
- The denominator, $Total\ Number\ of\ Pixels\ in\ Validation\ Set$ is the total number of validation pixels, providing a measure of classification reliability.

For 2004-2013 and 2014-2023, the Landsat 7 composite is classified using urban2013 and 2003 GCPs for non-urban classes, adapting to recent changes while retraining the RF model to reflect urban expansion.

3.4.2 Area Computation

Post-classification, areas are quantified in square kilometers. For each period, the area A_c of class *(c)* is calculated as:

$$A_c = Pixel\ Count_c * (Pixel\ Resolution^2/1{,}000{,}000)$$

Where:

$Pixel\ Count_c$ is the number of pixels classified as class (c),

$Pixel\ Resolution$ = 30m (Landsat's spatial resolution),

$Pixel\ Resolution^2$ = 30*30=900m^2 and divided by 1,000,000 converts square meters to square kilometer, yielding 0.0009 km^2 per pixel.

Thus, in order to ensure consistency with the ROI's total area while establishing a baseline for trend analysis, the equation simplifies to:

$$A_c = Pixel\ Count_c * 0.0009km^2$$

3.4.3 Prediction of Land Cover (2024-2033)

Future land cover for 2024-2033 is predicted by extrapolating spectral and area trends based on the full historical dataset from 1994 to 2023, incorporating data points from 2003 (1994-2003 composite), 2013 (2004-2013 composite), and 2023 (2014-2023 composite). To simplify, the annual spectral change rate per band is computed across the 20-year span from 2003 to 2023:

$$\Delta S_{band} = \frac{S_{2023,band} - S_{2003,band}}{20}$$

Where:

- $S_{year,band}$ is the median value of the specified band (e.g., SR_B1) in the composite for the respective period, and
- 20 is the number of years from 2003 to 2023, providing a straightforward annual change rate.

A simulated 2033 composite is generated by applying this rate forward from 2023:

$$S_{2033} = S_{2023,band} + (\Delta S_{band} * 10)$$

Where:

- $\Delta S_{band} * 10$ is the total change over the 10 years from 2023 to 2033, assuming a linear trend.
- This composite is classified using the RF model trained on 2014-2023 data, reflecting the most recent spectral characteristics and training points.

For area prediction, a simplified linear growth rate is calculated based on the total change from 2003 to 2033 as:

$$r_c = \frac{A_{c,2023} - A_{c,2003}}{20}$$

Where:

- $A_{c,2003}$ and $A_{c,2023}$ are areas of class (c) from the 1994-2003 and 2014-2023 classifications, and
- 20 is the number of years, giving the annual area change in km2.

The predicted area for class (c) in year (t) (2024-2033) is then computed as:

$$A_{c,t} = A_{c,2023} + \left(r_c * (t - 2023)\right)$$

Where:

- $A_{c,2023}$ is the area in 2023,
- r_c is the annual area change, and

- $t - 2023$ is the number of years from 2023 to (t) (for instance 1 for 2024, 10 for 2033), providing a linear year-by-year projection to 2033.

3.5 Evaluation Metrics

The performance of the Random Forest classifier is evaluated using a confusion matrix, a widely used metric in classification tasks to assess accuracy and error rates. In this project, the confusion matrix is generated within GEE to validate the classifications for each period (1994-2003, 2004-2013, 2014-2023) against ground truth data from GCPs. It provides a tabular summary of predicted versus actual class labels, enabling the calculation of overall accuracy and per-class performance.

Definition: For the four land cover classes (Urban [0], Bare [1], Water [2], Vegetation [3]), the confusion matrix is a 4x4 table where rows represent actual classes (from validation GCPs) and columns represent predicted classes (from the RF classifier). Each cell C_{ij} contains the number of pixels where the actual class is (i) and the predicted class is (j). Diagonal elements (C_{ii}) are correct predictions (true positives), while the off-diagonal elements are misclassifications. This is represented by:

3.5.1 Overall Accuracy

$$Overall\ Accuracy = \frac{\sum_{i}^{3} C_{ii}}{\sum_{i=0}^{3} \sum_{j=0}^{3} C_{ij}}$$

Where:

- $C_{ii}$is the number of true positives for class (i) – diagonal elements, and
- $\sum_{i=0}^{3} \sum_{j=0}^{3} C_{ij}$ is the total number of pixels in the validation set (sum of all matrix elements).

3.5.2 Producer's Accuracy (PA)

PA measures the proportion of actual pixels in a class correctly classified, reflecting the classifier;s ability to identify all instances of that class (omission error inverse).

$$PA_i = \frac{C_{ii}}{\sum_{j=0}^{3} C_{ij}}$$

Where:

- C_{ii} is the number of true positives for class (i), and

- $\sum_{j=0}^{3} C_{ij}$ is the total number of pixels predicted as class (i) (row total)

3.5.3 User's Accuracy (UA)

UA measures the proportion of predicted pixels in a class that are correct, reflecting the reliability of the classifier's predictions for that class (commission error inverse), represented as:

$$UA_i = \frac{C_{ii}}{\sum_{j=0}^{3} C_{ij}}$$

Where:

- C_{ii} is the number of true positives for class (i), and
- $\sum_{j=0}^{3} C_{ij}$ is the total number of pixels predicted as class (i) (column total)

3.6 Visualizations and Further Processing

Initial visualizations such as classified maps for 1994-2003, 2004-2013, 2014-2023, and 2024-2033, plus time series charts (2003-2033), bar and pie charts, and confusion matrices are generated in GEE. For further processing and enhanced mapping, data is exported from GEE to Google Drive. Classified images are saved as GeoTIFFs (e.g., Collection_1994_2003.tif), preserving the 30 m resolution and class values (0-3). GCPs are exported as Shapefiles (e.g., GCPs_2003.shp) for reference, and area statistics and chart data are saved as CSVs. Further processing to generate visualization map files are done using QGIS.

3.6.1 Software and Applications

This study employs a combination of software tools and programming languages to process satellite imagery, classify land cover, predict urban growth, and visualize results for Juba, South Sudan. The primary platforms used are Google Earth Engine (GEE) and QGIS, with JavaScript serving as the programming language within GEE. These tools facilitate data acquisition, preprocessing, classification, and visualization, ensuring a comprehensive workflow for urban growth prediction.

3.6.2 Google Earth Engine (GEE)

GEE, as a cloud-based geospatial processing platform, is designed for large-scale environmental data analysis. It provides access to extensive satellite imagery archives, including Landsat 5 and

Landsat 7 data, used in this study. The platform supports data acquisition, preprocessing, classification using the Random Forest algorithm (selected for this study), and visualization. It also offers preprocessing capabilities such as cloud and shadow masking, band scaling, and composite creation. The built-in Random Forest classifier is used for land cover classification and prediction, and results are visualized through maps, time series charts, and accuracy assessments. The GEE Code Editor, accessible via https://code.earthengine.google.com, serves as the development environment where the project script was written and managed, with integration to Google Drive for exporting processed outputs.

3.6.3 JavaScript in GEE

JavaScript is the primary scripting language for implementing the methodology in GEE. It enables the definition of functions, data handling, and algorithm implementation. Custom functions are written for image preprocessing, cloud masking, classification, and area computation. The language facilitates the loading of Landsat imagery, filtering by date and region of interest, and managing training data as feature collections. JavaScript commands allow seamless export of classified images and tabular data to Google Drive, while the GEE Earth Engine API supports efficient geospatial processing.

3.6.4 Random Forest in GEE

The Random Forest (RF) classifier is integrated into GEE as part of its supervised classification tools under the ee.Classifier namespace. It is implemented using the SMILE (Statistical Machine Intelligence and Learning Engine) library and optimized for large-scale geospatial data analysis. The RF classifier is instantiated using JavaScript (ee.Classifier.smileRandomForest(50)), where 50 decision trees balance accuracy and efficiency. Training data, consisting of labeled ground control points (GCPs), is used to train the classifier, which is then applied to median composites for historical periods and the predicted 2033 dataset. Validation is performed using confusion matrices to assess classification accuracy, and results are exported for further analysis.

3.6.5 QGIS

QGIS is an open-source Geographic Information System (GIS) software used for advanced visualization and spatial analysis of the study's outputs. It enables the import of GeoTIFFs and

Shapefiles while maintaining spatial resolution and metadata. The platform provides tools for symbology customization, comparative time series analysis, and cartographic enhancements such as legends and scale bars. QGIS also supports supplementary processing, including CSV integration for area statistics and spatial data manipulation. Its robust visualization capabilities enhance interpretability, making it a valuable complement to GEE.

Integration and Workflow

The integration of GEE and QGIS ensures an efficient workflow for urban growth prediction. GEE handles cloud-based data processing, classification, and initial visualization, while QGIS refines and presents the results offline. JavaScript scripts orchestrate the RF classification process in GEE, exporting outputs to Google Drive for import into QGIS. This workflow leverages GEE's computational scalability and QGIS's visualization capabilities, providing a robust pipeline from raw satellite data to actionable urban planning insights.

Chapter 4: Results and Discussion

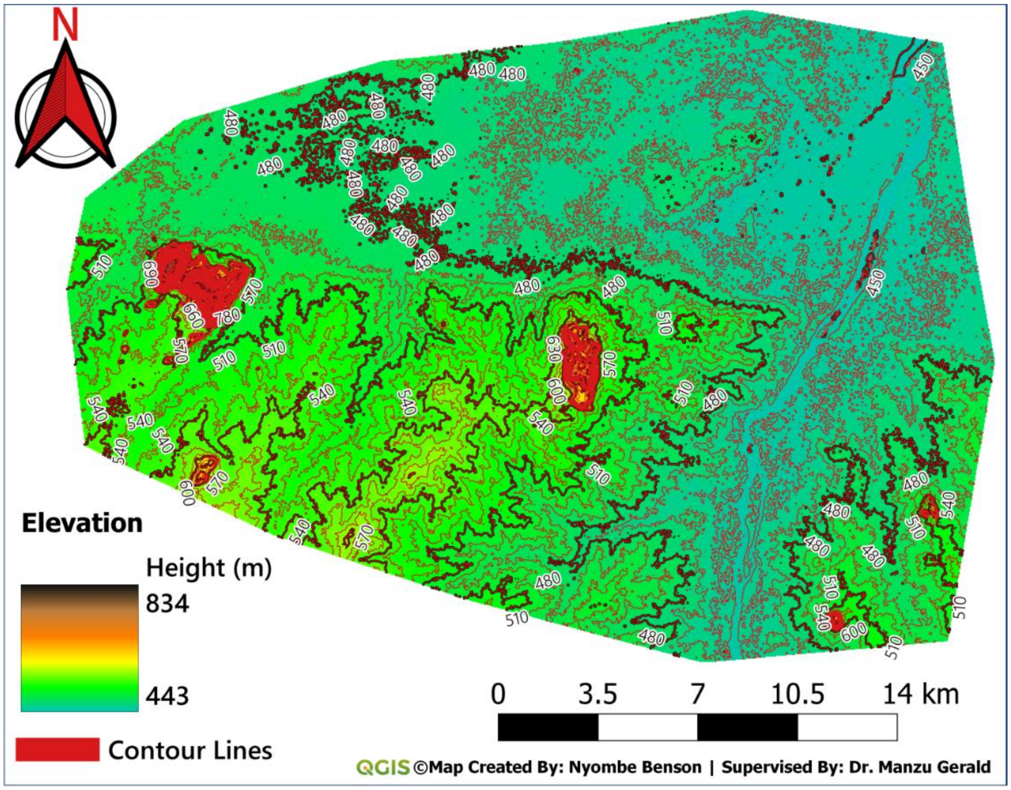
Chapter Summary

The results show that (1) built-up areas expanded dramatically, increasing from 9.44 km² (1.6%) in 1994–2003 to 54.72 km² (9.4%) in 2014–2023, with further growth projected to 81.93 km² (14.1%) by 2033 due to urbanization. (2) Bare ground fluctuated, peaking at 87.49 km² (15.1%) in 2004–2013 before declining, likely due to conversion to urban land. (3) Surface water remained stable (~1.9%) with minor variations, reflecting the White Nile's consistent influence. (4) Vegetative cover, initially dominant (483.24 km², 83.3%), declined to 450.73 km² (77.7%) by 2023 and is predicted to drop to 419.85 km² (72.4%) by 2033, signaling habitat loss. Urban growth followed infrastructure corridors, while topographic variations (430–800m elevation) shaped development patterns. The findings highlight urgent needs for sustainable planning to balance urbanization with environmental preservation, particularly in flood-prone areas near the White Nile.

The expansive study area surrounding Juba, covering approximately 580.08 square kilometers, has experienced transformations in land use and land cover patterns over the past three decades, with projections extending an additional decade into the future. With the Random Forest classifier, four primary land cover classifications were examined: built-up urban areas, exposed bare ground, surface water bodies, and vegetative cover. The temporal framework divides the analysis into three distinct historical periods (1994-2003, 2004-2013, and 2014-2023) along with a predicted future projection period (2024-2033).

4.1 Elevation of the Study Area

The study area has an elevation range of approximately 430 to 800 meters above sea level. Flowing from the south to the North through Juba, the White Nile plays a significant role in the area's topography. The lowest elevation is at the White Nile (Bahrl-al-Jebel) with an elevation of approximately 450 meters above sea level, while the highest point is further west of Juba, at Luri with an elevation of about 780 meters above sea level. The core of Juba, including neighborhoods like Gudele, Munuki, Kator and Jebel, les at around 470-510 meters above sea level, with higher areas towards the west or south east of the city (towards Yei and Nimule Highways, and Kajo-keji Road).



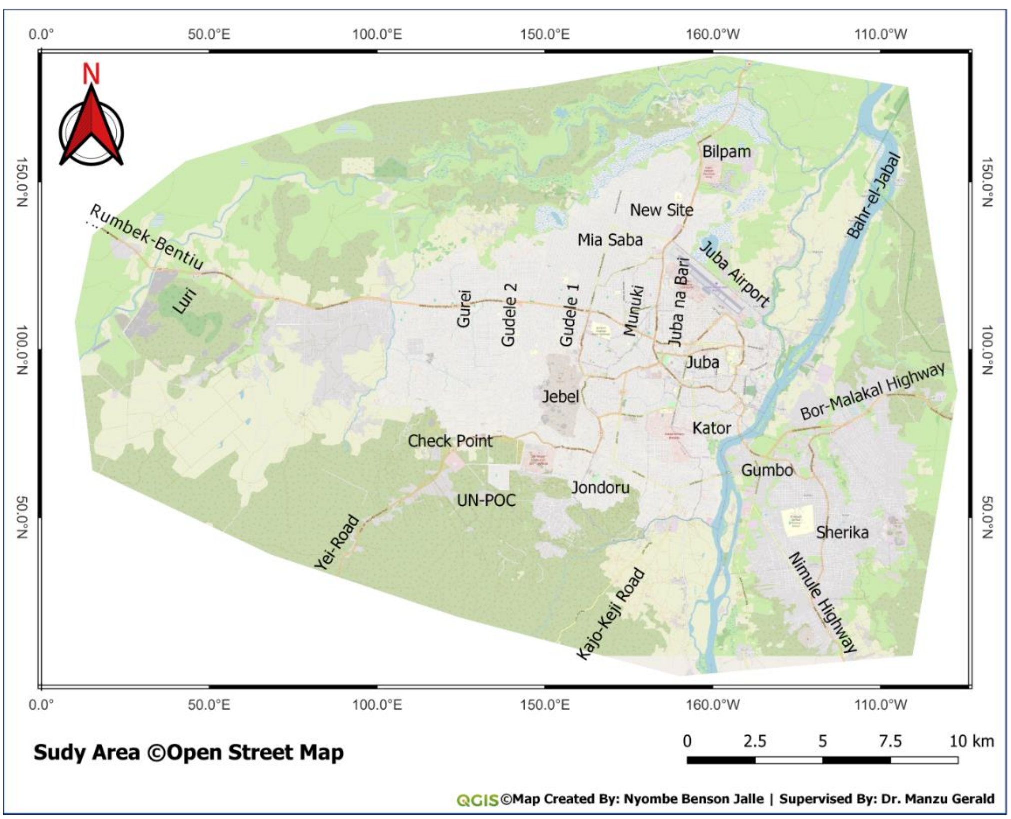


Figure 4-1: Elevation of the Study area

4.2 Analysis of land cover changes in the study area

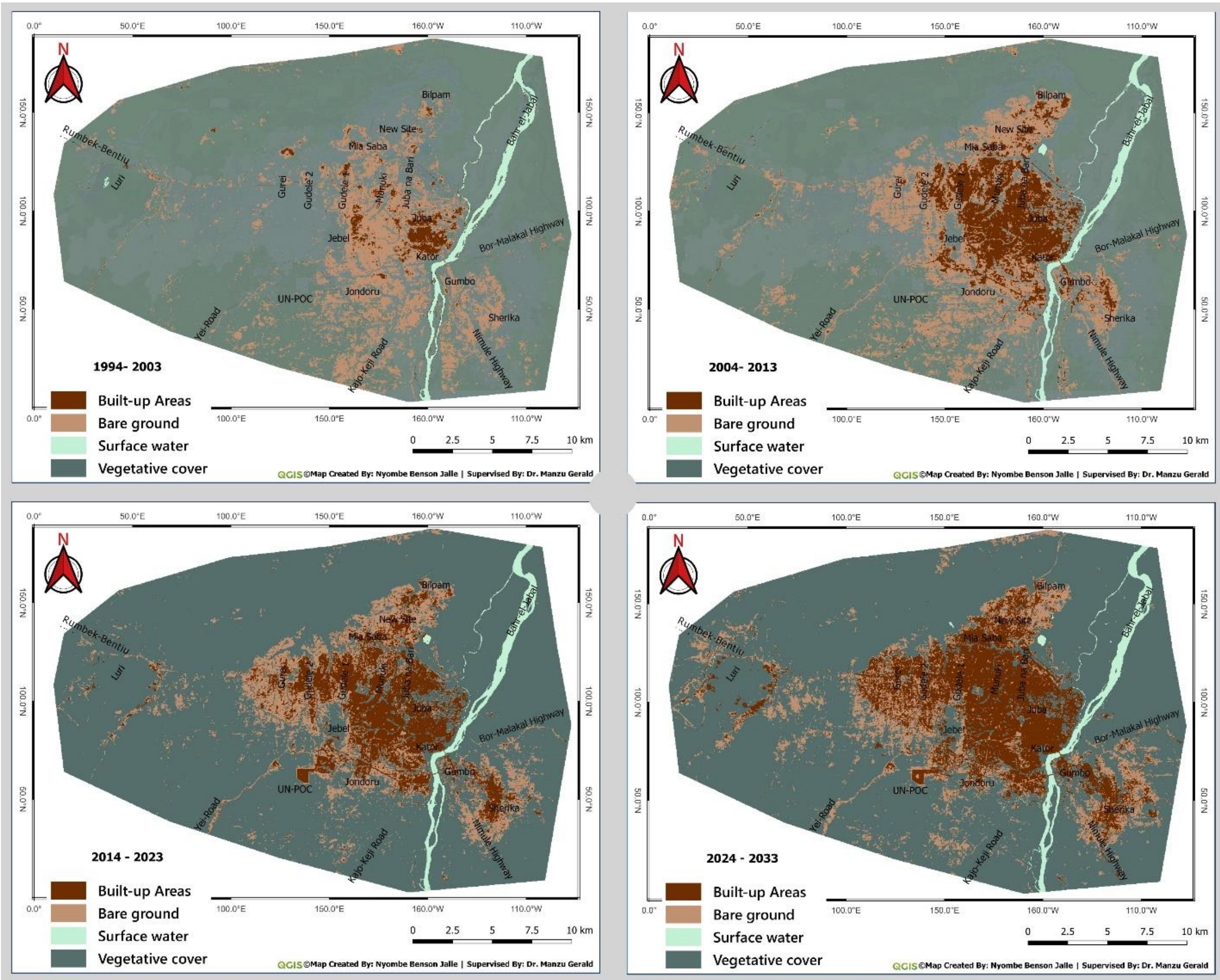


Figure 4-2: Prediction Analysis – time-series map (A)

4.2.1 Built-up Areas

The built-up area in the Juba region has shown a marked and consistent increase across the study periods. During the 1994-2003 period, the built-up area was 9.44 sq km, occupying just 1.6% of the 580.08 sq km study area.

By the 2004-2013 period, this area expanded significantly to 43.11 sq km (7.4%), reflecting rapid urbanization in Juba, likely driven by population growth and post-conflict development following South Sudan's Comprehensive Peace Agreement (CPA), which ended decades of conflict. The growth continued into the post-independence period, from 2014-2023, with the built-up area reaching 54.72 sq km (9.4%). The predicted built-up area for the 2024-2033 period is 81.93 sq km

(14.1%), suggesting continued urban development. Wider expansions of built-up areas followed patterns along infrastructure corridors.

Growth Rates: From the 1994-2003 period to the 2004-2013 period, the built-up area increased at an annual rate of 3.37 sq km/year over the 10-year interval, aligning with Juba's rapid development, from a state capital to an autonomous regional capital and finally to the country's capital city (July 9th 2011, onwards). From the 2004-2013 period to the 2014-2023 period, the annual growth rate slowed to 1.16 sq km/year, possibly due to a stabilization of urban expansion as available land becomes scarcer or development shifts to infilling – alternatively, it could draw inferences from a conflict that arose in 2013 just after independence, and again in 2016, which both did not spare the study area. The average annual growth rate from 1994 to 2023 is 2.26 sq km/year, indicating a sustained upward trend in urbanization over the three decades. For the predicted 2024-2033 period, the annual growth rate is expected to be 2.72 sq km/year, suggesting a resurgence in urban expansion.

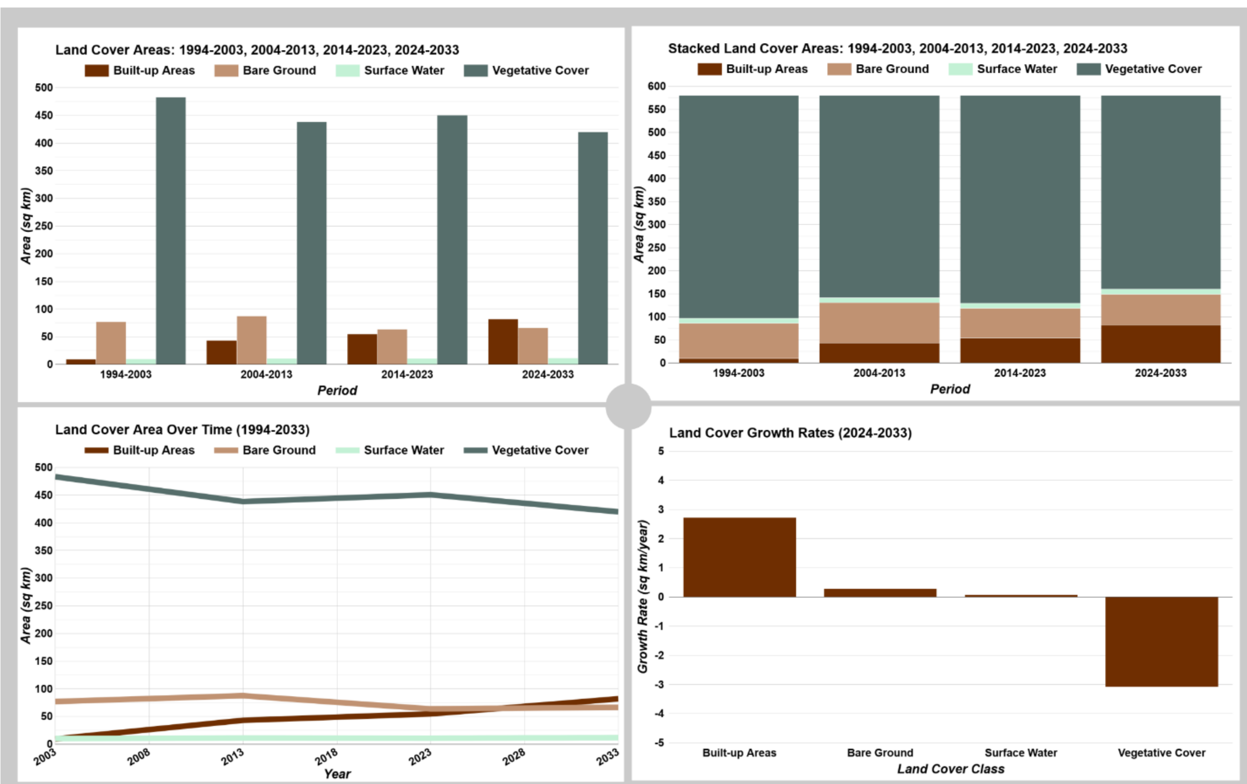


Figure 4-3: Prediction Analysis – timeseires 1994-2033 (B)

4.2.2 Bare Ground

Bare ground coverage displayed notable fluctuations over the study periods. In the 1994-2003 period, bare ground covered 77.12 sq km, accounting for 13.3% of the study area. This increased to 87.49 sq km (15.1%) by the 2004-2013 period, possibly due to land clearing for urban expansion, agriculture, or infrastructure development during this period of rapid growth. However, by the 2014-2023 period, bare ground decreased to 63.78 sq km (11.0%), possibly due to the conversion of bare ground into built-up areas, the regrowth of vegetative cover, or retraction of settlements from areas that were further expanding due to conflict(s). The prediction for the 2024-2033 period shows a slight increase to 66.63 sq km (11.5%), indicating a potential stabilization or minor expansion of bare ground.

Growth Rates: From the 1994-2003 period to the 2004-2013 period, bare ground increased at an annual rate of 1.04 sq km/year. However, from the 2004-2013 period to the 2014-2023 period, it decreased at a rate of -2.37 sq km/year, reflecting a significant reduction. For the predicted 2024-2033 period, bare ground is expected to grow at a rate of 0.29 sq km/year, suggesting a modest growth, possibly due to new land clearing activities as urban development continues.

4.2.3 Surface Water

Surface water, primarily associated with the White Nile (Bahr el Jebel), remained relatively stable with minor variations across the periods. In the 1994-2003 period, surface water covered 10.28 sq km, or 1.8% of the study area. This increased slightly to 11.13 sq km (1.9%) by the 2004-2013 period, possibly due to seasonal changes, improved detection in Landsat imagery, or minor changes in the river’s course. By the 2014-2023 period, surface water slightly decreased to 10.86 sq km (1.9%), potentially reflecting natural variability, or drier climatic conditions. The predicted surface water area for the 2024-2033 period is 11.67 sq km (2.0%), suggesting a potential rise in water coverage.

Growth Rates: From the 1994-2003 period to the 2004-2013 period, surface water increased at an annual rate of 0.09 sq km/year. From the 2004-2013 period to the 2014-2023 period, it decreased at a rate of -0.03 sq km/year. For the predicted 2024-2033 period, surface water is expected to increase at a rate of 0.08 sq km/year, indicating a slight rise in water coverage, possibly due to increased precipitation, or flooding events.

4.2.4 Vegetative Cover

Vegetative cover, the dominant land cover type in the study area, experienced a decline followed by a partial recovery, with a projected decrease in the future. During the 1994-2003 period, vegetative cover spanned 483.24 sq km, comprising 83.3% of the 580.08 sq km study area. By the 2004-2013 period, it decreased to 438.34 sq km (75.5%), likely due to urban expansion and land clearing for bare ground, particularly in areas further from the core city suburbs. By the 2014-2023 period, vegetative cover increased to 450.73 sq km (77.7%), indicating some natural regrowth or conflict related, as areas further from the core city became insecure. However, the prediction for the 2024-2033 period shows a decline to 419.85 sq km (72.4%), suggesting future pressure from continued urbanization and land use changes.

Growth Rates: From the 1994-2003 period to the 2004-2013 period, vegetative cover decreased at an annual rate of -4.49 sq km/year. From the 2004-2013 period to the 2014-2023 period, it increased at a rate of 1.24 sq km/year, suggesting natural regrowth, or seasonal patterns. For the predicted 2024-2033 period, vegetative cover is expected to decrease at a rate of -3.09 sq km/year, indicating future pressure from urbanization, potentially exacerbating habitat loss and environmental degradation.

4.2.5 Overall Trends and Implications

The land cover analysis reveals a clear trend of urbanization in the study area, with the built-up area increasing nearly ninefold from the 1994-2003 period to the projected 2024-2033 period, primarily at the expense of vegetative cover, which shows a net decline from 83.3% to 72.4% of the study area over the 40-year span. Bare ground peaks in the 2004-2013 period but decreases thereafter, while surface water remains relatively stable, with a slight projected increase by 2024-2033.

The growth rates are reflective of the dynamic interplay between land cover classes: built-up area consistently grows, with a notable resurgence in the 2024-2033 period (2.72 sq km/year), while vegetative cover faces significant declines (-3.09 sq km/year in 2024-2033). Bare ground shows variability, with a slight predicted increase (0.29 sq km/year), and surface water remains stable with minor growth (0.08 sq km/year). These changes align with Juba’s growth as a capital city, as evidenced by the expansion of neighborhoods further from the core, primarily three Payams of Juba City.

The predicted decline in vegetative cover raises concerns about habitat loss, soil erosion, and environmental degradation, particularly in a region with variations in topography (430-800 meters elevation) prone to flooding along the White Nile.

Chapter 5: Discussion and Conclusion

Chapter Summary

The Random Forest classifier demonstrated consistently high performance across all study periods, with overall accuracy ranging from 97.56% to 100% based on 82 validation pixels per period. The 1994-2003 period achieved perfect classification (100% accuracy) for all land cover classes (Urban, Bare, Water, and Vegetation). Accuracy slightly declined to 97.56% in 2004-2013, primarily due to bare ground misclassifications (PA/UA of 92.31%). Performance improved to 98.78% in 2014-2023 as bare ground classification accuracy increased to 96.15%. Urban, water, and vegetation classes maintained perfect accuracy (100% PA/UA) throughout all periods. The bare ground class showed the only classification challenges, with errors typically being misclassified as urban. The high overall accuracy scores (averaging 98.78%) validate the model's reliability for land cover analysis in this study. The consistent performance across three decades suggests robust classification methodology, though the bare ground class could benefit from additional training samples or spectral feature enhancements to further minimize its marginal misclassification rate.

5.1 Summary: Land Cover Transition and Predicted Growth Rates for the 2024-2033 Period in the Study area

Summary of Predicted Land Cover Transition (2014-2023 to 2024-2033)

The transition from the 2014-2023 period to the 2024-2033 period indicate significant changes in the distribution of land cover classes within the 580.08 sq km study area:

Built-up Area: The built-up area is predicted to increase from 54.72 sq km (9.4% of the study area) in 2014-2023 to 81.93 sq km (14.1%) in 2024-2033, a net gain of 27.21 sq km. The increase suggests that approximately 4.7% of the study area (27.21 sq km) will transition into built-up area, primarily at the expense of other land cover types.

Bare Ground: Bare ground is expected to rise slightly from 63.78 sq km (11.0%) in 2014-2023 to 66.63 sq km (11.5%) in 2024-2033, a net increase of 2.85 sq km. This modest gain of 0.5% of the study area indicates potential new land clearing activities, as a precursor to further urban development or agricultural expansion, though the overall change is relatively small.

Surface Water: Surface water is projected to increase by 0.1%, from 10.86 sq km (1.9%) in 2014-2023 to 11.67 sq km (2.0%) in 2024-2033, a net gain of 0.81 sq km.

Vegetative Cover: Vegetative cover, the dominant land cover type, is predicted to decrease from 450.73 sq km (77.7%) in 2014-2023 to 419.85 sq km (72.4%) in 2024-2033, a net loss of 30.88 sq km. The decline represents 5.3% of the study area, largely driven by the expansion of built-up area, with some vegetative cover likely converting to bare ground as well.

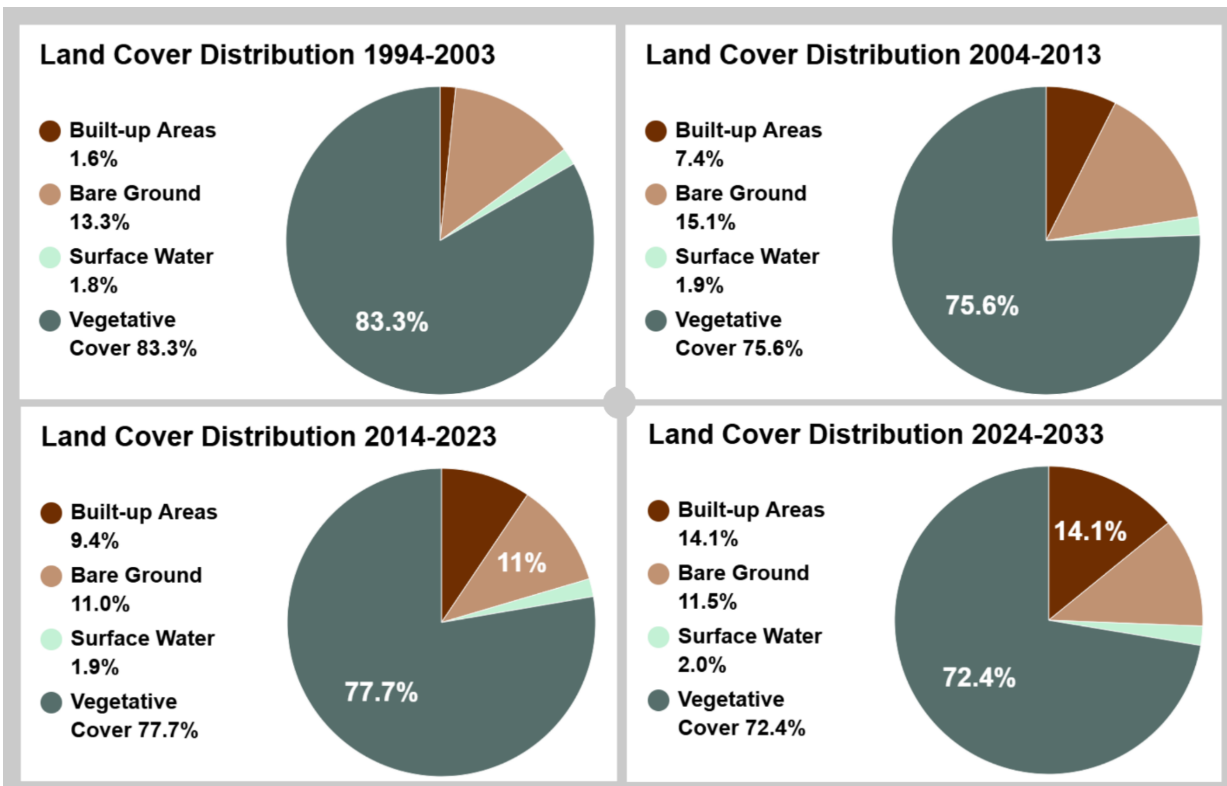


Figure 5-1: *Summary of changes*

Transition Dynamics: The primary transition in the 2024-2033 period is the conversion of vegetative cover to built-up area, with 27.21 sq km of the 30.88 sq km lost from vegetative cover likely being absorbed by urban expansion. The remaining 3.67 sq km of vegetative cover loss can be attributed to the combined gains in bare ground (2.85 sq km) and surface water (0.81 sq km). Bare ground and surface water show minor increases, suggesting that some areas may transition between these classes (e.g., bare ground near the White Nile or in the low-lying areas, becoming surface water during floods), but the dominant shift is from vegetative cover to built-up area, reflecting Juba's ongoing urbanization trend.

Predicted Growth Rates (2024-2033)

Built-up Area Growth Rate: The built-up area is expected to grow at an annual rate of 2.72 sq km/year, reflecting a significant increase from 54.72 sq km to 81.93 sq km, compared to the 1.16 sq km/year observed from 2004-2013 to 2014-2023.

Bare Ground Growth Rate: Bare ground is predicted to increase at an annual rate of 0.29 sq km/year, with a total gain of 2.85 sq km over the decade, though the rate is relatively low compared to historical fluctuations.

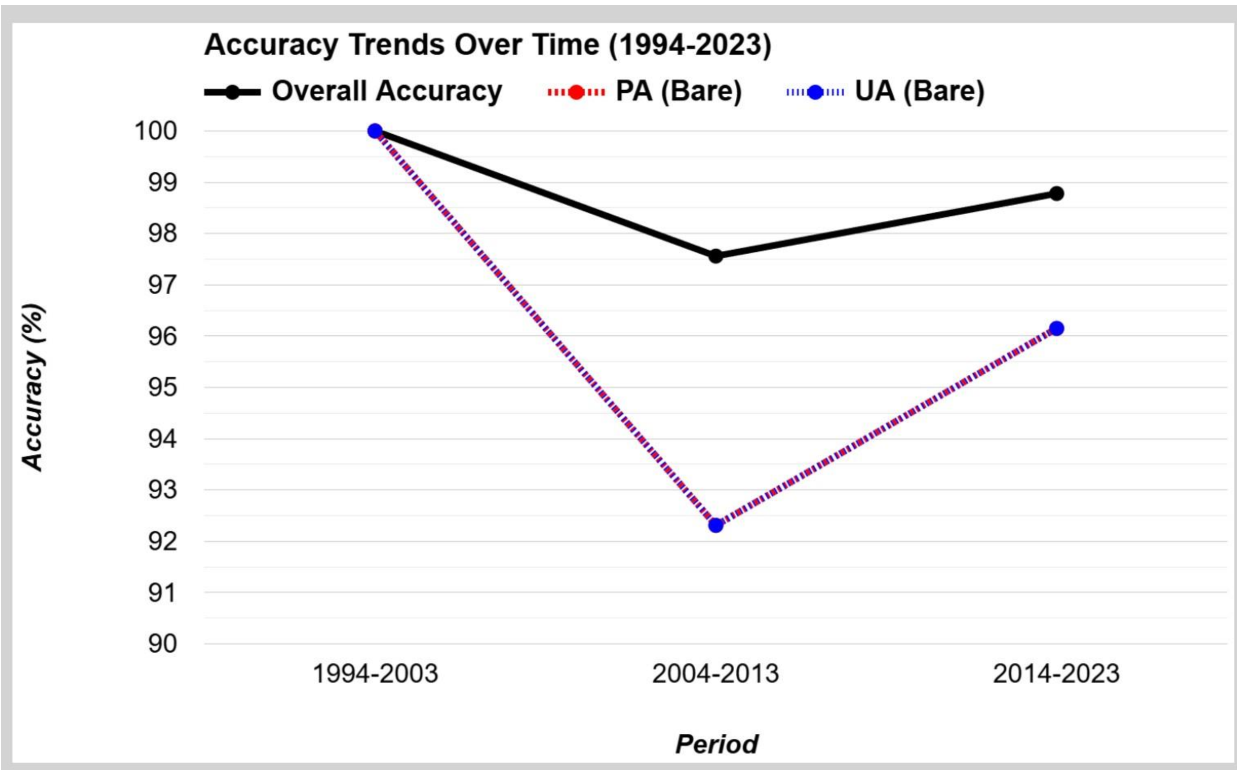
Surface Water Growth Rate: Surface water is expected to grow at an annual rate of 0.08 sq km/year, with a total increase of 0.81 sq km, with the White Nile maintaining its relatively stable presence in the landscape.

Vegetative Cover Growth Rate: Vegetative cover is predicted to decline at an annual rate of -3.09 sq km/year, with a total loss of 30.88 sq km over the decade potentially due to pressure from urbanization, as vegetative cover is converted into built-up area and, to a lesser extent, bare ground, posing risks to habitat and environmental stability.

5.2 Model Accuracy

The accuracy assessment results for the land cover classification across the three periods—1994-2003, 2004-2013, and 2014-2023—reveal a consistently high level of performance with minor variations, which can be effectively visualized using a line chart to highlight trends over time. For the 1994-2003 period, the classifier achieved a perfect overall accuracy of 100%, with both Producer's Accuracy (PA) and User's Accuracy (UA) at 100% for all classes: Urban, Bare, Water, and Vegetation. This flawless performance, based on a total of 82 validation pixels, indicates that the classifier, training data, and validation data were perfectly aligned during this period, resulting in no misclassifications across all land cover classes. In the 2004-2013 period, the overall accuracy slightly decreased to 97.56%, reflecting a small number of misclassifications among the 82 validation pixels used. The PA values remained at 100% for Urban, Water, and Vegetation, but dropped to 92.31% for Bare, indicating that some Bare pixels were not correctly classified, while the UA for Bare was also 92.31%, with the other classes maintaining 100%, as shown in a line chart where the Bare class's PA and UA dip noticeably compared to the flat 100% lines for the other classes.

For the 2014-2023 period, the overall accuracy improved to 98.78%, showing a slight enhancement over the previous period with the same 82 validation pixels. The PA values were 100% for Urban, Water, and Vegetation, and improved to 96.15% for Bare, indicating fewer misclassifications



compared to 2004-2013, while the UA values followed a similar pattern, with Bare at 96.15% and the other classes at 100%, a trend that a line chart would illustrate with Bare's PA and UA rising from 92.31% to 96.15%, alongside the overall accuracy's recovery from 97.56% to 98.78%. Overall, the classifier demonstrates robust performance across all periods, with overall accuracies ranging from 97.56% to 100% based on 82 validation pixels per period. The Urban, Water, and Vegetation classes were consistently classified with perfect accuracy (PA and UA of 100%), as would be evident in a line chart where their lines remain flat at 100%, while the Bare class presented the only challenge, with misclassifications as Urban in the 2004-2013 and 2014-2023 periods, reducing its PA and UA to 92.31% and 96.15%, respectively. The line chart would clearly show this trend, with Bare's PA and UA dipping in 2004-2013 and then improving in 2014-2023, suggesting potential enhancements in the training data, classifier, or image quality over time, and to further improve performance, investigating the spectral overlap between Bare and Urban, adding more training samples for Bare, incorporating features like NDVI, using period-specific validation data, and balancing the training data could help mitigate misclassifications in future analyses.

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